Lab 2

Responsive Navbar and Grid Drill

Detail	Value
Student name	Haseeb Alam
Roll number	SP24-BCS-142
Course	Web Development
Framework	Bootstrap 5.3.3 (CSS only, local copy) + vanilla JavaScript for the hamburger
Submission folder	lab-2/

1. Overview

This lab is a small rehearsal for Assignment 2. It has two parts: a responsive navbar (logo on the left, links in the centre, cart icon on the right, collapsing into a hamburger menu below 768px) and a product grid that shows 4 columns on desktop, 2 on tablet and 1 on phone. The content is placeholder text and product photos, because the point of the lab is the responsive mechanics.

2. How It Works

2.1 Breakpoint behaviour

Width	Navbar	Grid	Verified
375px (phone)	Hamburger + cart; links collapsed	1 column	Yes
767px	Hamburger (still collapsed)	1 column	Yes
768px (tablet)	Full navbar; links centred	2 columns	Yes
991px	Full navbar	2 columns	Yes
992px+	Full navbar	4 columns	Yes
1280px (desktop)	Full navbar	4 columns	Yes

Each row was checked in a real browser (Chromium) by measuring the toggler visibility, the number of cards per row and the horizontal position of the links. There was no horizontal scrolling at any width, no broken images and no console errors, and the hamburger menu was tested at 375px: it opens and closes on click, closes with the Escape key or after choosing a link, and resets when the window is resized to tablet width.

2.2 Key code

```
<nav class="navbar navbar-expand-md">
  <a class="navbar-brand site-logo">SAPPHIRE</a>
  <div class="nav-actions ... order-md-2">  cart + toggler  </div>
  <div class="collapse navbar-collapse order-md-1">
```

```
    <ul class="navbar-nav mx-auto"> ... </ul>
  </div>
</nav>
```

```
// js/script.js (vanilla JavaScript)
toggler.addEventListener("click", function () {
  setMenu(!menu.classList.contains("show"));
});
```

```
<div class="row row-cols-1 row-cols-md-2 row-cols-lg-4 g-4"> ... </div>
```

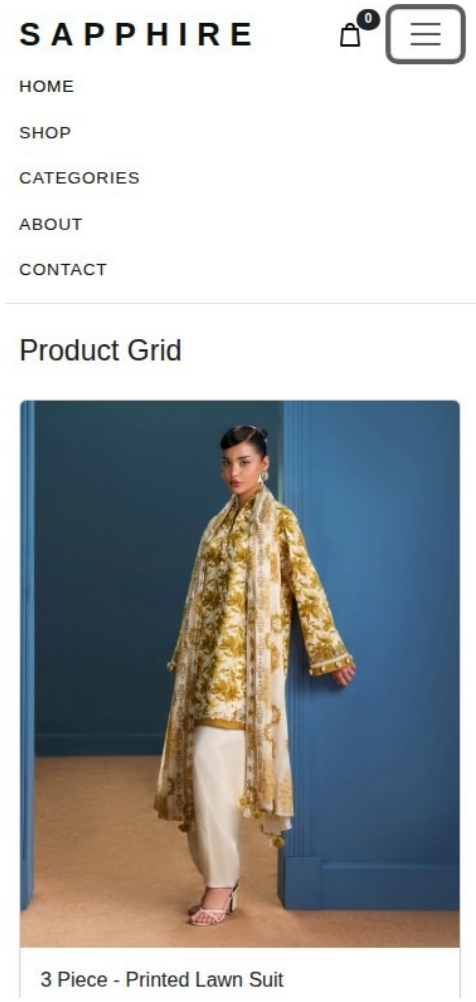
The cart and toggler come before the collapsing links in the HTML, so on a phone the collapsed menu drops to its own row below them. The order-md-* classes put the links between the logo and the cart from 768px upward. The JavaScript toggles the show class on the collapse container (the Bootstrap CSS then shows or hides it) and updates aria-expanded. Both HTML and CSS were validated with the Nu HTML Checker with no errors or warnings.

3. Screenshots

3.1 Phone (375px)



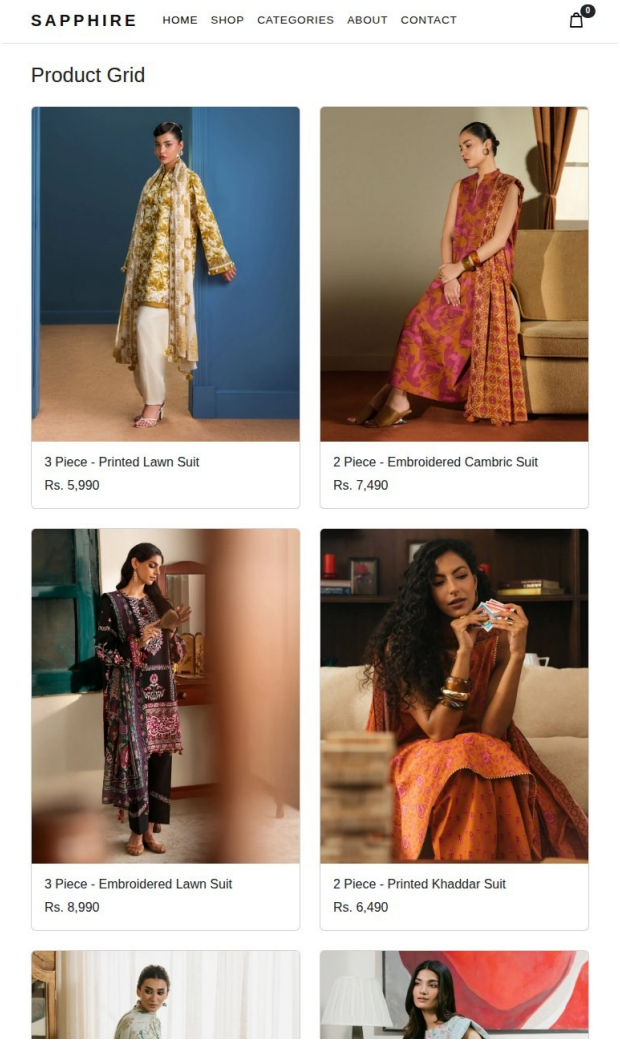
Hamburger menu closed



Hamburger menu open

Below 768px the grid is a single column, with each card full width.

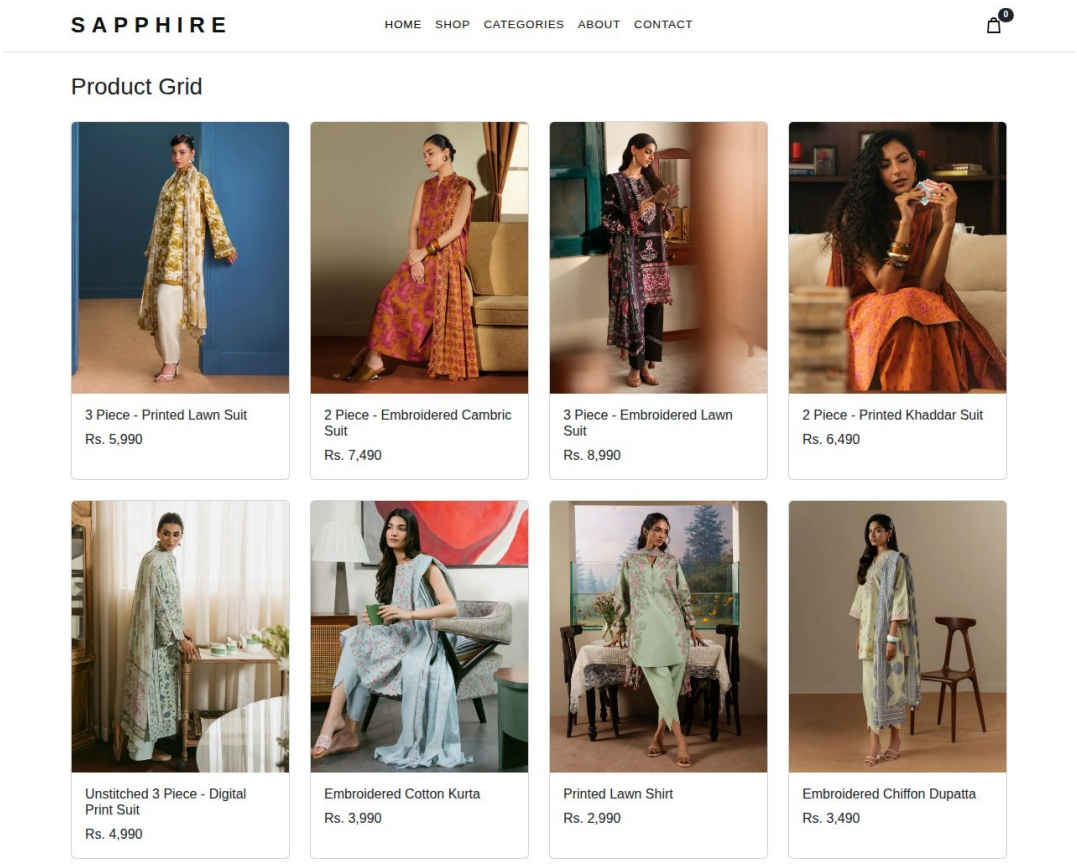
3.2 Tablet (768px)



Full navbar with centred links; 2-column grid

The top of the tablet page is shown; the full-length screenshots of every width are in the screenshots/ folder.

3.3 Desktop (1280px)



Full navbar; 4-column grid